Vortex Breakdown in Swirling Fuel Injector Flows

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It is well known that the process of vortex breakdown plays an important role in establishing the near-field aerodynamic characteristics of fuel injectors, influencing fuel/air mixing and flame stability. The precise nature of the vortex breakdown can take on several forms, which have been shown in previous papers to include both a precessing vortex core (PVC) and the appearance of multiple helical vortices formed in the swirl stream shear layer. The unsteady dynamics of these particular features can play an important role in combustion induced oscillations. The present paper reports an experimental investigation, using particle image velocimetry (PIV) and hot-wire anemometry, to document variations in the relative strength of PVC and helical vortex patterns as the configuration of a generic fuel injector is altered. Examples of geometric changes that have been investigated include: the combination of an annular swirl stream with and without a central jet; variation in geometric details of the swirler passage, e.g., alteration in the swirler entry slots to change swirl number, and variations in the area ratio of the swirler passage. The results show that these geometric variations can influence: the axial location of the origin of the helical vortices (from inside to outside the fuel injector), and the strength of the PVC. For example, in a configuration with no central jet (swirl number $S=0.72$), the helical vortex pattern was much less coherent, but the PVC was much stronger than when a central jet was present. These changes modify the magnitude of the turbulence energy in the fuel injector near field dramatically, and hence have an important influence on fuel air mixing patterns. [DOI: 10.1115/1.2799530]

Introduction

Although vortex breakdown was first observed in a study of delta wing leading edge aerodynamics (Peckham and Atkinson [1], the relevance to the highly swirling flows produced by gas turbine combustion system fuel injectors soon became apparent (e.g., Gupta et al. [2]). The term vortex breakdown refers to: "*an abrupt and drastic change in flow structure which is characterised by a sudden axial deceleration that occurs above a certain level of swirl, leading to the formation of a free stagnation point followed by a separation region with turbulence behind it*," Wang and Rusak [3]. Because of the complex nature of the phenomenon and its significance for several important engineering applications, it has received considerable attention in the literature for the last 50 years, covering experimental, theoretical, and computational investigations. Two comprehensive reviews have been produced by Escudier [4] and Lucca-Negro and O'Doherty [5]. Nevertheless, there is still no general theory of vortex breakdown or agreement on its fundamental nature.

Experimentally, Sarpkaya [6] and Leibovich [7] discovered seven distinct types of vortex breakdown; however, in turbulent flows only two, the spiral and bubble modes, were observed. Sarpkaya [6] also showed that the effect of an adverse axial pressure gradient was to move the vortex breakdown further upstream, or if extreme enough, even to initiate it. In addition to the appearance of a recirculation region in the bubble mode, evidence has been gathered that vortex breakdown can display strongly time-dependent characteristics, whereby the recirculation zone itself becomes unstable and starts to precess about its own axis (a precessing vortex core (PVC), Syred and Beer [8]). Finally, an important concept that appears in most explanations of vortex breakdown is the idea of "criticality" (Benjamin [9]), i.e., whether the flow axial velocity is, throughout the flow, greater than the phase velocity of longitudinal inertial instability waves (supercritical), or less than this (subcritical). In the latter case, a mechanism exists whereby the downstream conditions may affect the upstream flow. Indeed, perhaps a useful analogy for the occurrence of vortex breakdown is that it allows a supercritical flow to transition to a subcritical one, so that it may then gradually adjust to downstream conditions, in the same way that a supersonic flow requires a shock wave to be formed, so that a subsonic flow is created that can then sense the downstream pressure field. It is for this reason that care has to be taken in both experimental and numerical studies of vortex breakdown to pay attention to the region of influence of downstream conditions. Escudier et al. [10] have recently shown the importance of this for highly swirling flows downstream of a nozzle or orifice.

The significance of vortex breakdown for the design of gas turbine fuel injectors that employ high swirl for flame stabilization reasons is well known. Li and Gutmark [11], in measurements on a triple annular swirler typical of future fuel lean module injectors, have shown that the dynamics of vortex breakdown cause interactions between vortical structures in the flow associated with breakdown and acoustical instability modes. The measurements show that the vortical structures are highly dependent on swirler geometry. The fundamentally unsteady nature of vortex breakdown and its association with instability modes of the flow has attracted the attention of computational fluid dynamics (CFD) model developers, particularly using large eddy simulation (LES). Studies of highly swirling flow in confined (Wegner et al. [12]) and unconfined (Garcia-Villalba and Froehlich [13]) geometries have recently been reported. The former noted the appearance of what was described as "*a double-cell precessing vortex core*," but it is not clear whether this phrase is mixing up the appearance of rotating spiral vortex structures and a genuine PVC. The latter study observed that the introduction of a central pilot jet inside an annular swirl stream had little influence on the average flow, but spectra extracted from the LES predictions indicated dramatic effects on the instantaneous spiral vortex structures. This result is very significant, but it needs to be confirmed that it is correct for

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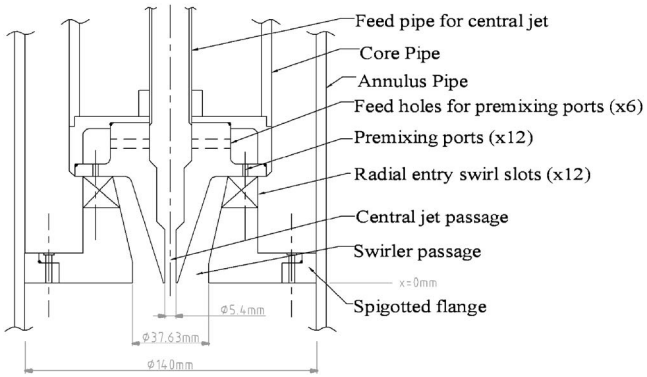

Fig. 1 Datum fuel injector geometry

more practically relevant confined swirl flow. Finally, in a detailed experimental study that included both a central jet and an annular swirl stream, Midgley et al. [14,15] deduced from PIV measurements that multiple spiral vortex structures were observed wrapped around a central recirculation zone (CRZ). This should perhaps be interpreted as a vortex breakdown in which both spiral and bubble modes occurred simultaneously. The origin of the spiraling vortex structures was traced to a separation event inside the fuel injector, and the "two strong/two-weak" double helix form of the spiral vortices was explained by the need for the secondary flowfield of the vortex structures to satisfy continuity within the annulus. As noted in other works, the near fuel injector turbulence field was strongly influenced by the appearance of the spiraling vortex structures. Based on the above, the measurements of Midgley et al. [15] have been extended to explore the influence of a central jet in a confined flow, and the effect of the swirl and swirl passage geometry on vortex breakdown.

Test Facility

The test rig used for the present measurements is the same as in the study reported by Midgley et al. [15]; details are given in Refs. [14,15]. Tests were conducted in both water and air flow, but superior PIV data were obtained in water flow and these are reported here, with hot wire data from airflow tests. The radially fed swirler (12×30 deg angled slots) produced an annular swirlstream (outer diameter $D_s=37.63$ mm) at fuel injector exit (Fig. 1). The inner diameter of the annulus was formed by the outer wall of a central jet nozzle (inner diameter $D_j=5.4$ mm). The standard test condition corresponded to swirler and jet Reynolds numbers of 8×10^4 and 2.63×10^4 defined using D_s and D_j and the bulk mean velocities based on geometrical exit areas: $V_{x,j}=4.88$ m/s, $V_{x,s}=2.13$ m/s. Note, however, that the discharge coefficient of the swirl passage was measured at 0.5, so the peak swirlstream velocity was about 1.5 $V_{x,s}$. In the data reported in [15], the central jet had always been present; the current experiments concentrated initially on the case with no central jet, but an unchanged swirlstream condition. To allow further exploration of changes in the swirlstream conditions, a modular injector was designed (Fig. 2). This allowed changes to the radial swirler slot angle (α_1), and the angles of the central and outer body (α_2,α_3) to assess the effects of swirl passage shape. In all tests with the modular injector, no central jet was present. Note also that the annular swirl passage in the modular geometry did not contain an internal corner, as present in the datum geometry. This was intentional to allow assessment of whether the presence of this internal corner in the datum geometry was a "trigger" to the internal vortex breakdown, as suggested in [15]. By selecting $\alpha_2=17$ deg and $\alpha_3=9.5$ deg, (Case 1) the injector exit plane bulk axial and swirl velocities of the swirlstream were the same in datum and modular fuel injectors. By varying (α_2,α_3), the bulk average axial velocity ratio of the swirl passage flow between outlet and inlet

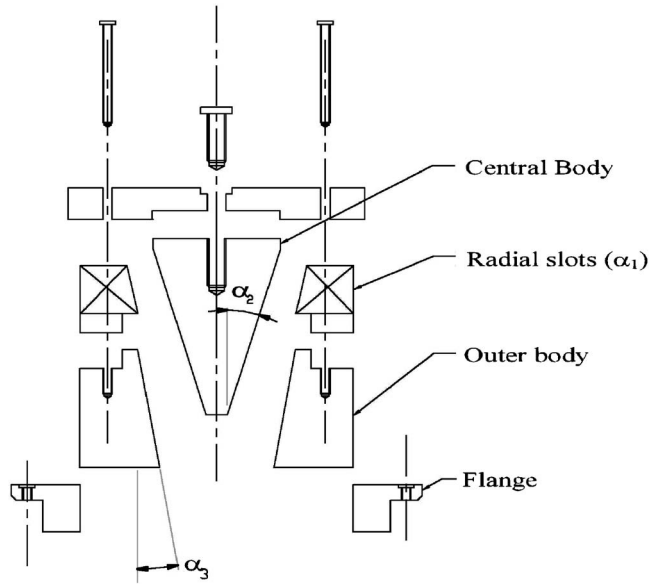

Fig. 2 Modular fuel injector geometry

($V_{x,\mathrm{so}}/V_{x,\mathrm{si}}$), and similarly the swirl velocity ratio ($V_{t,\mathrm{so}}/V_{t,\mathrm{si}}$), could be altered to investigate the effect of different levels of acceleration and "spin up" within the swirlstream passage on the flow produced. Table 1 shows the various cases considered. Case 6 used central body and outer body modules of constant radius (14/25 mm) to provide a parallel swirl passage configuration. Finally, to assess the impact of the swirl number on the flow, different radial slot modules were constructed with various values of α_1. Table 2 shows the resulting swirl numbers as deduced from exit plane PIV measurements.

Instrumentation

The PIV instrument comprised a LaVision system with a twin Nd: YAG laser generating a 1 mm thick light sheet via spherical and cylindrical lenses, and a Kodak Megaplus ES 1.0 camera with 1008×976 pixel resolution, running at 15 Hz (a few measurements at 1 kHz were also made with an upgraded camera/PIV system). Laser/camera synchronization, image acquisition, and data processing using the cross-correlation technique were controlled by Davis software running on a dual processor Pentium III PC. 20 μm polyamide particles (almost neutrally buoyant in water) were used as light scattering particles. The PIV instrumenta-

Table 1 Modular injector geometry parameters

	α_2 (deg)	α_3 (deg)	$V_{x,\mathrm{so}}/V_{x,\mathrm{si}}$	$V_{t,\mathrm{so}}/V_{t,\mathrm{si}}$
Case 1	17	9.5	1.25	1.56
Case 2	20.5	8.5	1.3	1.56
Case 3	13.5	11.5	1.43	1.56
Case 4	13.5	8.5	1.29	1.65
Case 5	20.5	11.5	1.21	1.43
Case 6	0	0	1.0	1.0

Table 2 Radial swirl module slot angles and swirl numbers

α_1 (deg)	S
30	0.72
20	0.57
10	0.47
0	0

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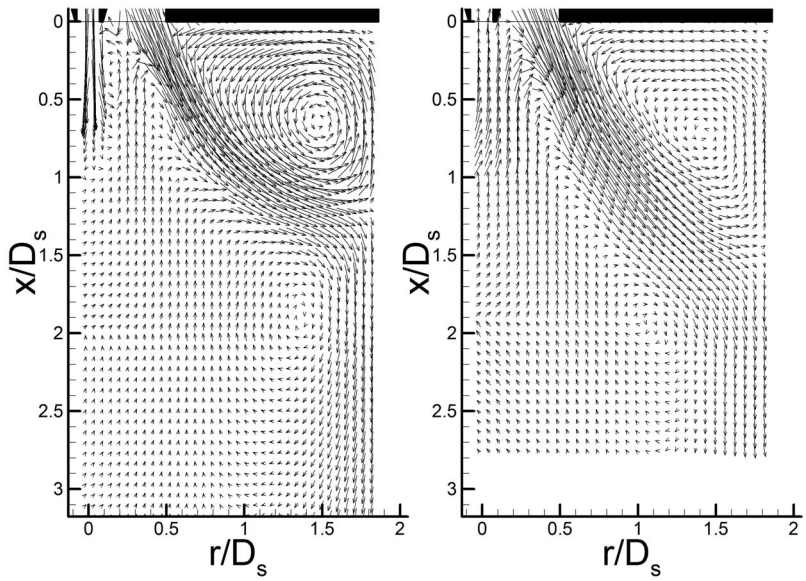

Fig. 3 Overall flow structure; datum injector (a) with fuel jet; (b) without fuel jet

tion parameter setup was selected following the recommendations of Keane and Adrian [16]. Laser power and optical system/data collection parameters were optimized by minimizing the interframe time Δt to reduce truncation errors in velocity evaluation, but not setting Δt so small as to incur precision errors in particle location due to subpixel interpolation. The interframe time (particle shift) was set so that the inplane displacement was less than $1/4$ of the PIV interrogation cell used for cross correlation and, where possible, led to out-of-plane motion less than $\delta z/4$, where δz is the camera depth of field. The optimum particle image diameter is around 2 pixels. For 20 μm particles, this was achieved in the present experiment when the field of view (FoV) was smaller than 80×80 mm^2. In most data collected, there were at least ten particle pairs per interrogation window. The interrogation cell used a deformed multipass grid (LaVision [17]) of initial size 64×64 pixels, reducing to a final size of 32×32 pixels (with 50% overlap). The interframe delay time was around 500 μs for the larger FoV images used for initial identification of flow structures (160×160 mm^2) and 20–50 μs for the smallest FoV (20×20 mm^2). This minimized out-of-plane data loss and maximized the inplane displacement, which was typically from 3–8 pixels. The same approach to data validation was applied as described in [15]; a range of FoVs was used that allowed optimization of PIV system parameters to the length scale of local dynamically important flow structures. The local turbulence length scales were estimated using the technique developed by Spencer and Hollis [18], and the measured turbulence levels corrected (increased) using the formula proposed in [18] if the size of the PIV interrogation cell was not small enough, relative to the local integral length scale. In all the data presented here for statistical quantities, time averaging over 650 PIV frames has been carried out.

Results and Discussion

Comparison of Flowfield With and Without Central Jet. Figure 3 presents the mean velocity vectors in the *x-r* plane measured in the datum injector geometry. In comparison with the overall flow structure for the case with a central jet shown in Fig. 3(*a*) [15], the no jet case in Fig. 3(*b*) shows that the central recirculation zone (CRZ) extends right up to the injector exit plane, the corner recirculation is weaker and less well defined, with the swirl cone attachment point on the outer wall further downstream ($x/D_s=1.5$ compared to the value $x/D_s=1.2$, reported in [15]). The absence of the central jet changes the turbulence structure in the centerline region in the first 2–3 swirler diameters dramatically. Figure 4 compares measured radial RMS turbulence inten-

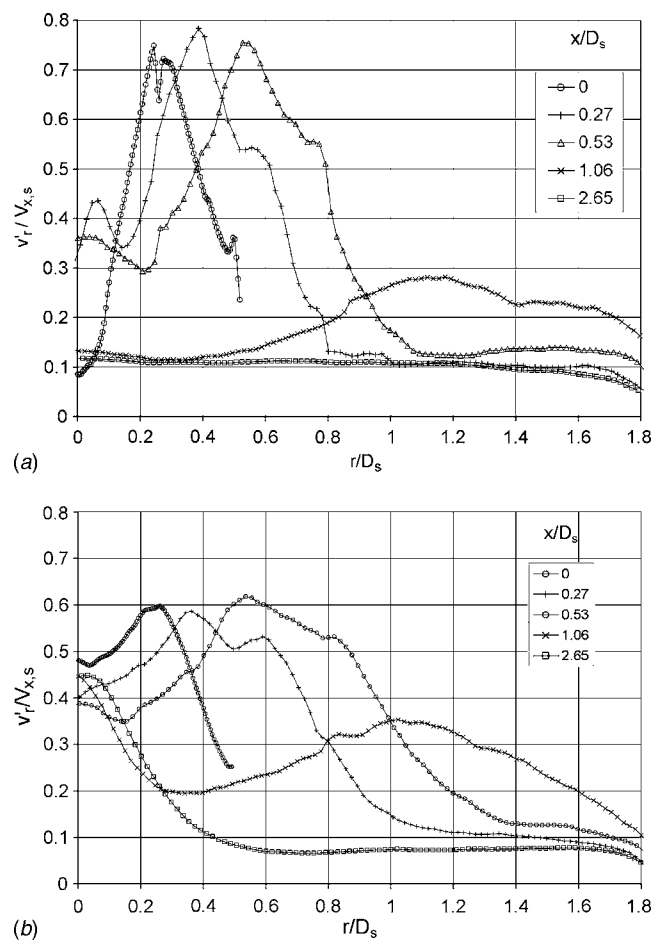

Fig. 4 Radial RMS profiles; (top) with central jet; (bottom) no central jet

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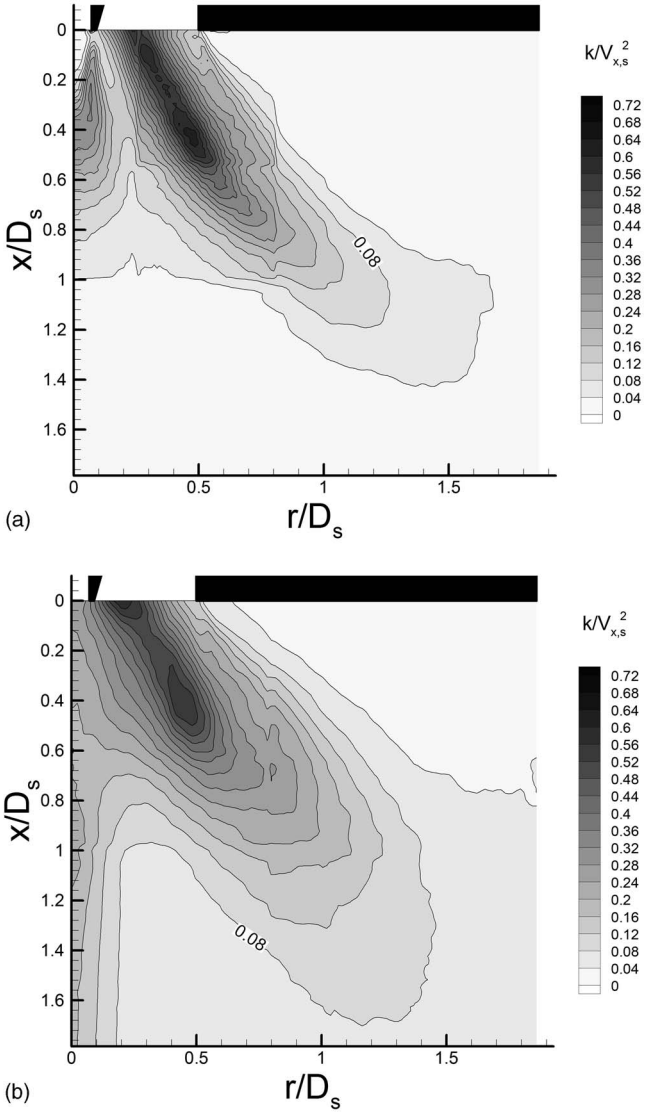

Fig. 5 Turbulence kinetic energy contours; (top) with central jet; (bottom) no central jet

sity profiles in this region for flows with and without a central jet (similar changes are observed in axial and circumferential RMS data). Outboard of $r/D_s \sim 0.3$, the results are quite similar in the two flows. A striking difference is observed, however, closer to the centerline. With a central jet, the turbulence levels start low, rise to a peak at $x/D_s \sim 0.5$ and then decay rapidly beyond this. This structure is connected to the high axial velocity gradients in the region of $x/D_s = 0.5-1.0$ associated with the free stagnation point between forward flowing central jet and backflow in the CRZ. Of course with no central jet, as noted above, the CRZ reaches back to the injector exit plane. The turbulence levels in the no-jet case are quite different. Figure 4 shows that these start high and remain very high ($\sim 45\%$) right down to $x/D_s \sim 3$. Comparison with measured axial and circumferential turbulence intensities (Midgley [19]), indicates that the radial and circumferential turbulence levels are significantly higher (a factor of ~ 2) compared to the axial data. This is recognizable as an effect of the unsteady fluctuations associated with a PVC (that would influence radial and tangential motions more than axial), which are picked up by the time-averaging process as "turbulence." The implication is that the presence of a central jet stabilizes the CRZ and damps its tendency to precess; more measurements to confirm that the no central jet case does have a strong PVC will be presented below. Further illustration of the changed fluctuating velocity field brought about in the near-injector region by the omission of the central jet is provided in Fig. 5, which compares the measured turbulence kinetic energy fields. In both cases, the peak turbulence levels are associated initially with the inner edge of the swirl-stream shear layer (and identified in [15] as caused by the rotating multiple vortex system). For no central jet, however, the maximum value is lower (0.6 compared to 0.75). The swirl shear layer is clearly more diffused in the no jet case, and the streak of high fluctuations measured on the centerline and stretching downstream is again the contribution from the PVC, which is clearly stronger in the no jet case. It is important to note again here that, for the high values measured in the shear layer edge, and on the centerline in the no jet case, these should not be considered as genuine turbulence, since a large contribution is due to high energy periodic events. A further piece of evidence that shows the no-jet case is markedly more unstable than the central jet case is contained in Fig. 6; this shows the PDF of the outer wall attachment point of the swirl cone. The larger variance in this ($x/D_s = 0.5-2.5$) than when a central jet is present ($x/D_s = 0.75-1.75$)

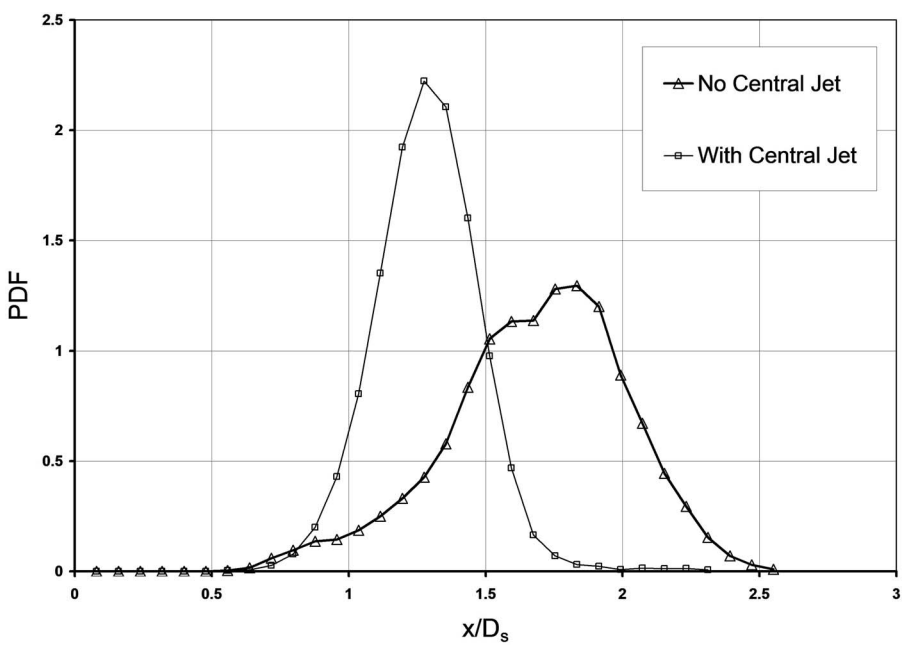

Fig. 6 PDF of spatial attachment point of swirl cone

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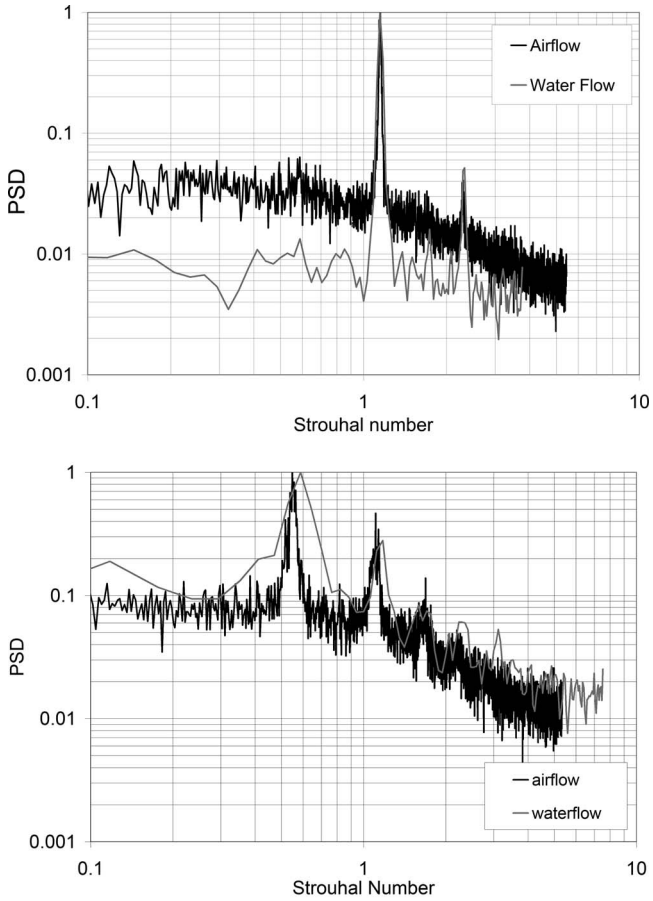

Fig. 7 PSD of axial velocity in swirlstream shear layer (top) with central jet; (bottom) no central jet

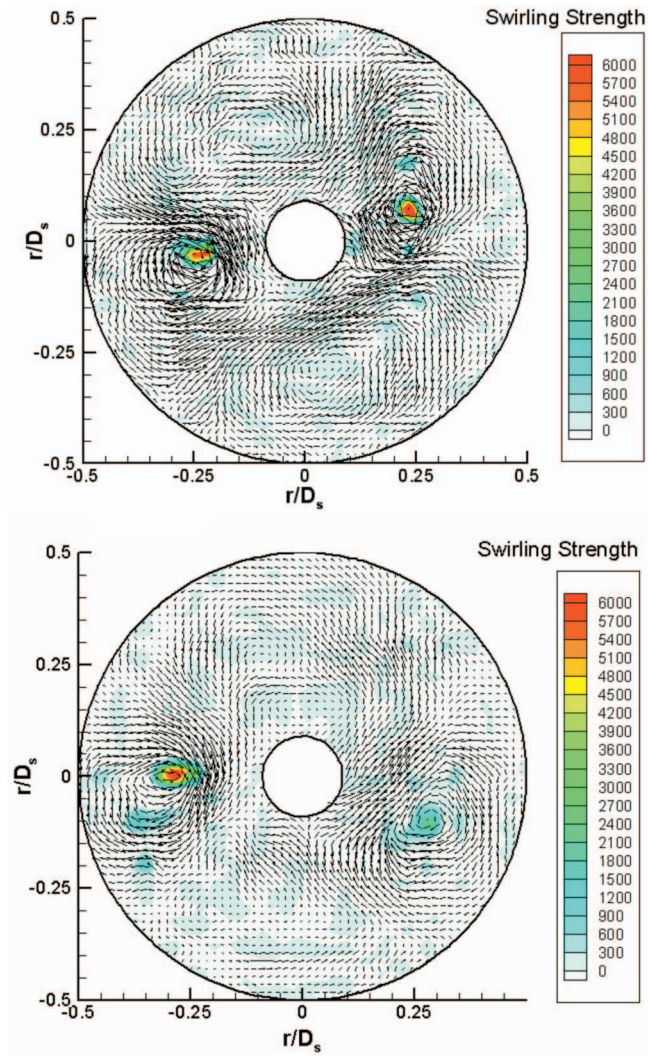

Fig. 8 Conditionally averaged velocity vectors in $r-\theta$ plane at $x/D_s=0.0$; (top) with central jet; (bottom) no central jet

indicates a more vigorous "flapping" of the swirl stream cone, once again probably induced by the strong PVC. Finally, Fig. 7 compares power spectral density (PSD) measurements of the axial velocity taken at a point within the inner swirlstream shear layer edge at injector exit ($x/D_s=0.27$, $r/D_s=0.27$, $\theta=0$ deg). These data were taken in an airflow experiment with a hot wire at 10 kHz resolution, and also in the water flow experiment with high speed PIV (1 kHz); good agreement between the two experimental techniques is seen.

Strongly periodic events are identifiable in both spectra. In the flow with a central jet, the Strouhal numbers at which these occur (St=1.32 and 2.71) were identified in [15], with the rotation of a two strong/two weak double vortex pair system originating with a breakdown event inside the fuel injector. With no central jet, the largest peaks appear at Strouhal numbers of 0.6 and 1.2. The energy is also distributed over a broader range of frequencies near these peaks. The appearance of multiple peaks confirms that a vortex pair system must still be present in the no jet case, although probably weaker as was seen in the reduced turbulence energy maximum value in Fig. 5. Following the technique described in [15], conditionally averaged vectors were derived for the two cases at injector exit and are shown in Fig. 8, superimposed on a swirl strength parameter. The two strong/two weak vortex pattern is very obvious in the central jet data; with no central jet, only two vortices can be clearly identified, and these are also much less coherent with quite unequal swirl strength parameters. As a consequence of this, the main periodic feature identified as these structures rotate is the passage of the strongest vortex past the measurement point, with a weaker signal from the passage of the vortex pair (since these are less coherent), and a much weaker signal from the hardly visible second vortex pair. This explains why the Strouhal peaks in the no jet case are at half the value of the with jet data.

Further, an additional periodic event is now present in the no jet case that was not present with a jet, namely the PVC. In order to extract quantitative details on the PVC, the decision was taken to look further downstream (at $x/D_s=2.65$), where evidence for a PVC had first been seen in the turbulence statistics. Instantaneous PIV data of velocity vectors in the $r-\theta$ plane are shown in Fig. 9 at two instants of time. The aerodynamic center of the swirling flow can be easily identified, and is clearly precessing in time about the geometric center (identified by the black dot). The variation in time of the angular location of the aerodynamic center was taken from the instantaneous PIV data and is shown in Fig. 10. The sawtooth pattern indicates a coherent rotational motion at the very low frequency of around 0.75 Hz, a Strouhal number of only ~0.01, as defined in the above data using swirl passage exit parameters. If the Strouhal number is redefined (St_{duct}) using the downstream duct diameter (140 mm) as a length scale and the associated bulk mean velocity, then St_{duct} takes on a value of ~0.7, which is in good agreement with the data of Syred [20] on PVC characteristics at a bulk swirl number of around 0.75, as in the present case.

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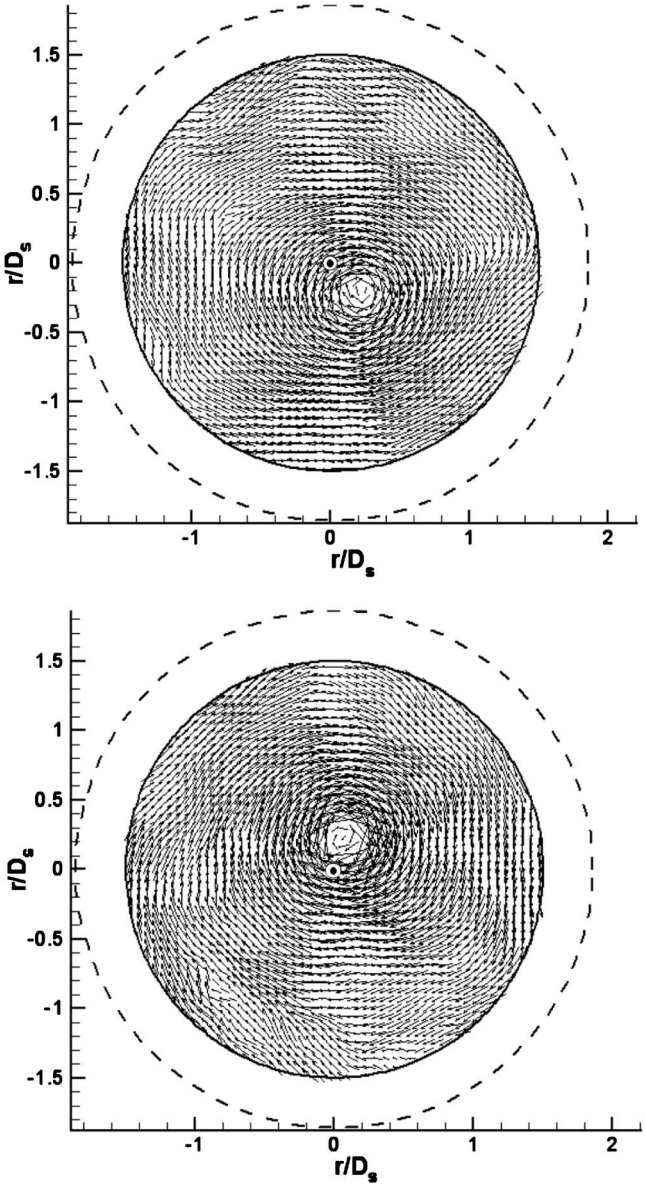

Fig. 9 Instantaneous velocity vectors in $r-\theta$ plane at x/D_s =2.65, no central jet

Modular Injector Geometry Data. The first data taken with the modular injector were for Case 1, which was set up to be the same as the no jet flow case discussed above, except for the removal of the internal corner. Figure 11 presents a series of spectra measured at $x/D_s=0.27$ with a hot wire in airflow for Case 1 at 25 kHz resolution with swirl slots at $\alpha_1=30$ deg. Two spikes in the spectra are observed at the same Strouhal numbers as in the datum injector geometry at all radial locations measured. This result shows conclusively that the separation event inside the injector identified in [15] as the origin of the double vortex pair was not associated with the internal corner. By reducing the swirl slot angle (α_1), and hence swirl number, the change in spectral characteristics may be observed in Fig. 12. The intensity of the dominant spikes reduces with swirl level until at $\alpha_1=10$ deg, where no spikes are observed. Although the data are shown here for the Case 1 swirl passage geometry, the same result was obtained for all Cases studied. The swirl number clearly affects the appearance of the internal vortex breakdown, creating the multiple vortex structures, but the passage geometry has no discernible influence.

Measurements reported in [15] with the datum injector had suggested that the origin of the vortex structures was inside the injector. Using the modular injector geometry, the improved optical access allowed this to be confirmed, see, Fig. 13, where the mean separation point is measured as $x/D_s=-0.25$. This separation point is also mainly determined by the swirl number, moving downstream to $x/D_s=-0.1$ for $\alpha_1=20$ deg, and by $\alpha_1=10$ deg there is no internal separation (Fig. 14). This unequivocally links

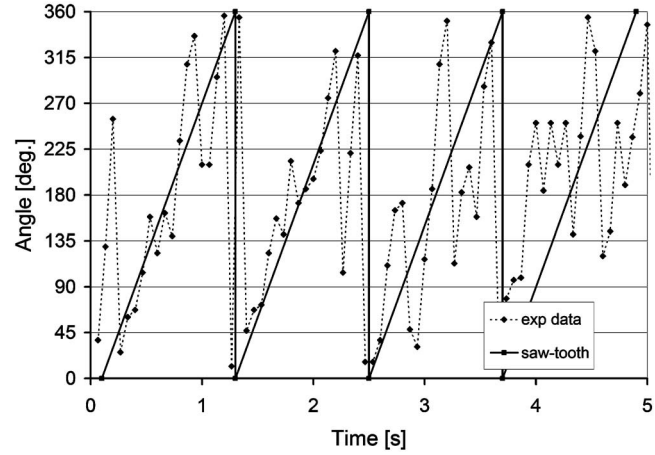

Fig. 10 Angular location of swirl pattern aerodynamic center at x/D_s=2.65, no central jet

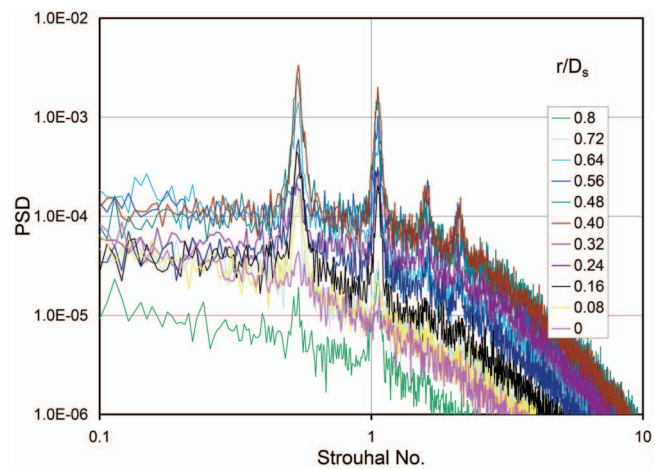

Fig. 11 PSD in modular injector, Case 1, x/D_s=0.27; α_1 =30 deg

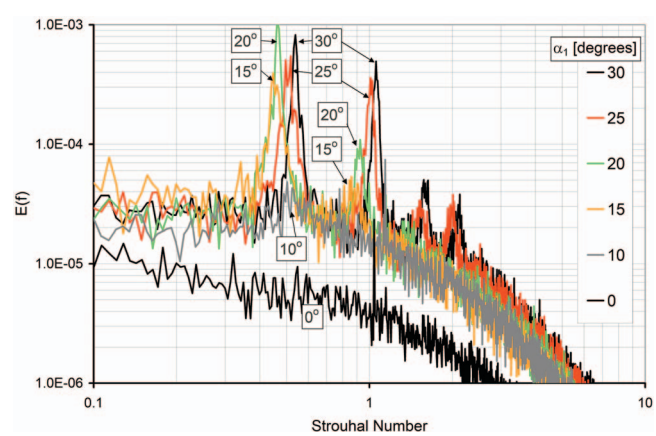

Fig. 12 PSD in modular injector, Case 1; various slot angles

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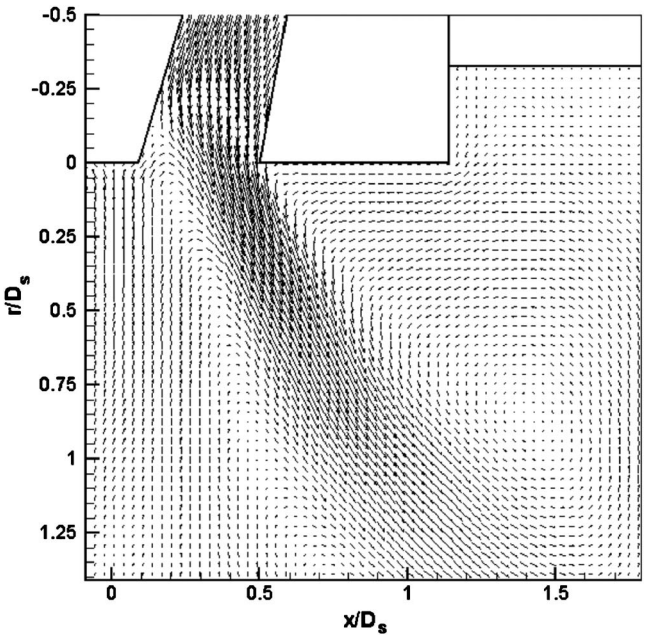

Fig. 13 Velocity vectors, modular injector, Case 1; α_1=30 deg

the internal flow separation and the multiple spiraling vortex structures. As a final piece of evidence, Fig. 15 shows the measured velocity vectors for Case 6 with a parallel walled swirl passage, superimposed on the turbulence energy contours. For α_1=30 deg and a parallel passage, the resulting swirl number was higher than used before at 0.95. However, there is now no separation inside the injector and the peak turbulence level at 0.35 at the injector exit plane is half that observed previously. From hot wire data taken at x/D_s=0.27, providing the PSD data shown in Fig. 16, evidence was found of spikes in the spectra but for this geometry only for $\alpha_1 > 20$ deg. The implication is that the spiraling vortices have still appeared at high enough swirl numbers, but their origin is now outside the injector. The most likely explanation is that the formation point of the vortices is determined by an interaction between the forward flow in the swirlstream passage and the backflow occurring in the CRZ, with the swirl number being the dominant parameter controlling this, and the swirl passage geometry playing an indirect role, since changes here will influence the precise swirl number at injector exit.

Conclusions

A detailed experimental study has been reported of the near-field injector flowfields created as various flow and geometric parameters of the injector were varied. The main emphasis was on the effect of the presence or absence of a central nonswirling jet in addition to the annular swirlstream. The absence of the jet was shown to lead to large destabilization of the flow. The CRZ began to precess creating a PVC, and large levels of velocity fluctuations were measured as a consequence on the centerline for a significant region downstream of the injector exit plane. The PVC unsteadiness influenced the precise nature of the vortex breakdown, which was confirmed to occur inside the injector for the baseline flow conditions studies. The swirler cone itself also became more unstable without a central jet, producing a large increase in the variation in time of its attachment location on the confining geometry outer wall. Geometry variations in the swirl passage were examined but had only minor influence on the flow characteristics, except for their influence on the injector exit swirl number, which was the dominating parameter. Variations in the swirl number were shown to lead always to a spiral form of vortex breakdown with multiple vortex structures wrapped around the CRZ. Depend-

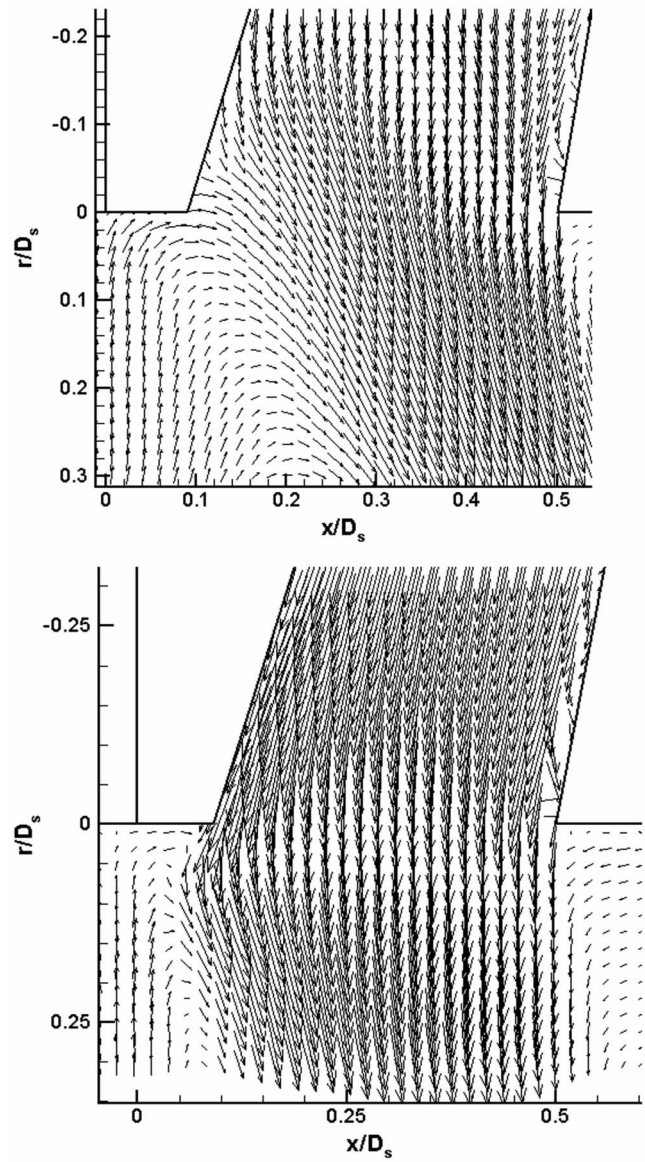

Fig. 14 Velocity vectors, modular injector, Case 1; (top) α_1 =20 deg; (bottom) α_1=10 deg

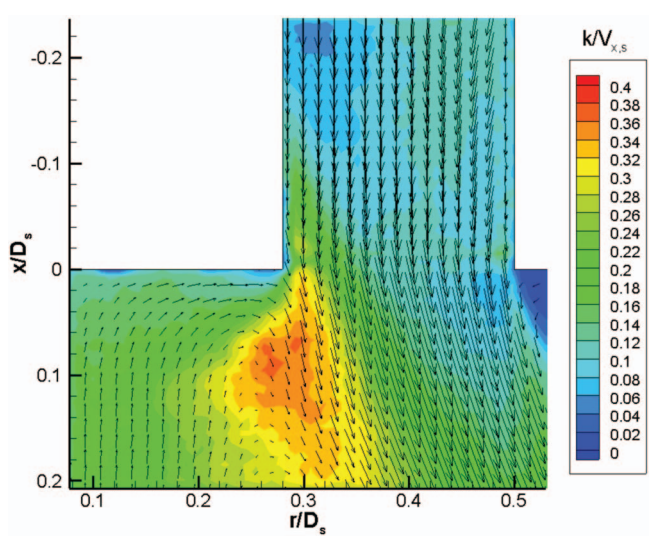

Fig. 15 Velocity vectors and *k* contours, modular injector, Case 6; α_1=30 deg

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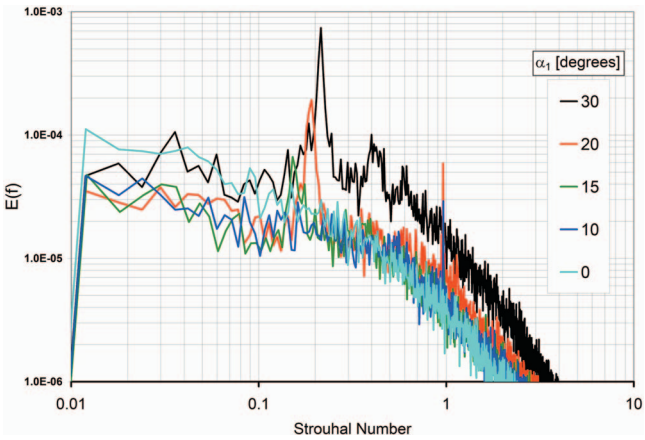

Fig. 16 PSD for modular injector, Case 6; various slot angles

ing on the geometry, the precise value of swirl number at which these spiraling structures appeared varied, as did the location of their first appearance, which could be both inside and outside the injector.

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Nomenclature

D = cylindrical combustor duct diameter
D_j = central jet diameter
D_s = annular swirler passage diameter
f = frequency of periodic peaks in PSD data
k = turbulence kinetic energy, or direction k
L_{ij}^k = length scale of i/j velocity fluctuation correlation in k direction
$\underline{r}$ = separation vector
$\overline{R_{ij}}$ = spatial correlation function
S = swirl number
St = dimensionless Strouhal number $(fD_s/V_{x,s})$
u_i, u_j = velocity fluctuations in i,j directions
V_x, V_r, V_t = time-mean velocities in x, r, θ directions
$V_{x,j}$ = bulk axial velocity of central jet stream
$V_{x,s}$ = bulk axial velocity of annular swirl stream
v_x', v_r', v_t' = fluctuating RMS in x, r, θ directions
x, r, θ = cylindrical polar coordinates
Δt = interframe time
δz = camera depth of field

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